Business Requirements Document (BRD)

Solar Service Management Mobile Application (Solar-App)

1. Executive Summary & Product Vision

1.1 Objective

The Solar Service Management Mobile Application is a comprehensive digital platform designed to streamline solar operations—from lead capturing and interactive solar estimation to customer onboarding, scheduled preventive maintenance (AMC), complaint tracking, and automated billing.

1.2 Target Audience & Core Personas

1. **End Customer:** Residential, society, or commercial solar installation owners requiring real-time tracking of generation metrics, automated service scheduling, complaint registration, and digital invoice clearance.
2. **Field Technician / Service Engineer:** On-ground technicians who receive ticket assignments, complete preventive maintenance checklists, capture site proof, and request spare parts.
3. **Sales & Survey Agent:** Field representatives who capture lead details, execute site feasibility assessments, and generate instant quotes.
4. **System Administrator:** Back-office operators responsible for ticket allocation, customer document verification, inventory management, and financial reporting.

2. Core Modules & Detailed Feature Matrix

Module 1: Authentication & User Profile Management

1.1 Mobile Number & OTP Authentication Screen

- **Functionality:** Secure passwordless access for customers and field agents using mobile OTP verification.
- **Feature Breakdown:**
    - **Country Code & Phone Input:** Standard 10-digit mobile number input with validation.
    - **OTP Dispatch & Resend Logic:** Automatic SMS trigger with a 60-second cooldown timer.
    - **Auto-Read OTP:** SMS auto-fill permission handling on Android devices.
    - **PIN/Biometric Quick Login:** Option to set a 4-digit PIN or enable Fingerprint/FaceID for subsequent app launches.
- **Business Rules:**
    - Unregistered mobile numbers will be redirected to the Lead Registration pipeline.
    - Max 3 invalid OTP attempts allowed per hour before temporary lockout (15 minutes).

1.2 User Profile & Role Switching Screen

- **Functionality:** Displays active user attributes, associated site details, and language preferences.
- **Feature Breakdown:**
    - **Multi-Property/Site Selector:** Switch between different installed sites (e.g., "Home - 5kW" vs "Factory - 50kW") under one customer account.
    - **Profile Editing:** Update secondary contact numbers, communication email, and emergency contacts.
    - **App Preferences:** Toggle language (English/Marathi/Hindi) and push notification settings.

Module 2: Interactive Solar Calculator & Lead Capture

2.1 Feasibility Calculator Screen

- **Functionality:** An interactive, instant estimation engine for potential leads and customers considering capacity expansion.
- **Feature Breakdown:**
    - **Monthly Electricity Bill Slider:** Interactive slider ranging from ₹500 to ₹1,000,000+.
    - **Rooftop Area Input:** Numerical input for available shade-free area (sq. ft. or sq. m.).
    - **Connection Type Selector:** Radio selection (Residential Individual, Housing Society, Commercial/Industrial).
    - **Instant Estimation Summary Card:**
        - Recommended Solar Capacity (in kW).
        - Estimated Project Cost (Gross).
        - Estimated Government Subsidy Amount (DBT calculation under PM Surya Ghar Scheme).
        - Net Investment Cost & Estimated Monthly / Annual Savings.
        - Payback Period (in years/months) and CO_2 Reduction Metrics.
- **Business Rules:**
    - Subsidy calculations must automatically apply current state/central slab rules (e.g., 3kW cap for maximum residential subsidy).
    - Area validation: Minimum 100 sq. ft. per 1 kW system required; display warning if rooftop area is insufficient for target bill reduction.

2.2 Instant Quote Generation & Lead Conversion Screen

- **Functionality:** Converts calculator estimates into a formal PDF quotation and captures lead parameters.
- **Feature Breakdown:**
    - **Lead Details Form:** Full Name, Installation Pincode, Preferred Contact Time.
    - **Schedule Feasibility Site Visit:** Date & Time slot picker for on-site assessment by a solar engineer.
    - **PDF Quote Preview & Download:** One-click generation of branded PDF quote with itemized component breakdown (Panels, Inverter, Structure, Balance of System).

Module 3: Customer Onboarding & Site KYC Management

3.1 Customer Profile Setup Screen

- **Functionality:** Captures detailed technical and legal attributes for sanctioning and installation.
- **Feature Breakdown:**
    - **Consumer Account Number Details:** Electricity Provider (DISCOM) selection, Consumer Number, Meter Number, Sanctioned Load (kW).
    - **Address & Geolocation Tagging:** Address fields with auto-capture of latitude/longitude coordinates via mobile GPS for technician routing.

3.2 Document Upload & Verification Screen

- **Functionality:** Upload required KYC and DISCOM approval documents.
- **Feature Breakdown:**
    - **Document Checklist:**
        1. Aadhaar Card / PAN Card (Identity Proof).
        2. Latest Electricity Bill (Address & Sanctioned Load Proof).
        3. Property Ownership Proof / NOC (Tax receipt or agreement).
        4. Roof Site Photos (Pre-installation).
    - **Camera & Gallery Integration:** Direct photo capture with automatic image compression (max 2MB per upload).
    - **Verification Status Badge:** Real-time badge indicating document review state (Pending Verification, Approved, Action Required / Rejected).
- **Business Rules:**
    - Rejection requires mandatory comment entry by admin (e.g., "Electricity bill unreadable, re-upload").

Module 4: Asset & Installation Tracking

4.1 Site Equipment Register Screen

- **Functionality:** Digital twin of installed solar equipment for warranty and servicing tracking.
- **Feature Breakdown:**
    - **Solar Panel Specs:** Brand, Model, Serial Numbers (Barcode/QR Code scanner input), Total Quantity, Individual Panel Wattage.
    - **Inverter Specs:** Inverter Serial Number, Capacity (kW), Phase Type (Single Phase / 3 Phase), Monitoring Dongle ID.
    - **Structure & Battery Details:** Mounting Structure Type (GI High-Rise, Rooftop Flush), Battery Bank details (if Hybrid/Off-grid).
    - **Warranty Card Vault:** Digital repository of panel linear performance warranties (25 years) and inverter warranties (5–10 years).

Module 5: Preventive Service Management & AMC Operations

5.1 Service Milestone Dashboard Screen

- **Functionality:** Tracks free service schedules and active Annual Maintenance Contracts (AMC).
- **Feature Breakdown:**
    - **Free Service Timeline:** Visual timeline showing the 4 mandatory free quarterly services post-commissioning.
    - **AMC Status Banner:** Active plan details (Basic Cleaning, Comprehensive Breakdown Included), Expiry Date, and Renewal CTA.
    - **Service History List:** Log of past completed services with downloadable Service Reports signed by field engineers.

5.2 Service Request & Booking Screen

- **Functionality:** Customer requests a routine cleaning or preventive health checkup.
- **Feature Breakdown:**
    - **Service Type Selection:** Routine Panel Cleaning, Complete System Inspection, Inverter Health Check.
    - **Preferred Slot Selector:** Select date and morning/afternoon time window.
    - **Technician Allocation Tracker:** Displays assigned technician name, photo, phone number, and real-time arrival status.
- **Business Rules:**
    - Free services must be claimed within 30 days of the due milestone date; otherwise, status updates to Lapsed.

Module 6: Complaint Ticketing & Breakdown Resolution

6.1 Raise Complaint Ticket Screen

- **Functionality:** Rapid escalation path for system faults, zero generation, or physical damage.
- **Feature Breakdown:**
    - **Issue Category Dropdown:** (Zero Generation / Inverter Red Light, Physical Panel Damage, Structure Loose / Water Leakage, Generation Lower Than Expected, Billing / App Connection Issue).
    - **Severity Indicator:** Low, Medium, High, Emergency (e.g., Sparking/Fire Risk).
    - **Problem Description & Media Attachment:** Text box + ability to attach up to 4 photos/videos of error codes on inverter screens.
    - **Auto-Ticket Number Generation:** Instant creation of unique tracking ID (e.g., TKT-2026-0892).

6.2 Ticket Lifecycle & Live Status Screen

- **Functionality:** End-to-end audit trail of complaint resolution.
- **Feature Breakdown:**
    - **Status Stepper:** Ticket Raised $\rightarrow$ Assigned to Engineer $\rightarrow$ Technician En-Route $\rightarrow$ On-Site Inspection $\rightarrow$ Parts Replaced $\rightarrow$ Resolved & Closed.
    - **In-App Communication:** Call or chat directly with assigned service engineer.
    - **Customer OTP Resolution Verification:** Customer must provide a 4-digit OTP to the technician on-site to officially mark the ticket as Resolved.

Module 7: Field Technician Workspace (Mobile Workflow)

7.1 Job Queue & Task List Screen

- **Functionality:** Daily dashboard for field engineers showing assigned routine services and emergency breakdown tickets.
- **Feature Breakdown:**
    - **Map Route Optimization:** Overview map displaying all assigned site visits ordered by distance/time priority.
    - **Job Acceptance:** Accept/Reschedule job assignments with reason capture.

7.2 Service Execution & Inspection Checklist Screen

- **Functionality:** Digital job sheet completed on-site before sign-off.
- **Feature Breakdown:**
    - **Mandatory Checklist Items:**
        - Panel VOC / ISC Voltage measurement checks.
        - AC/DC Distribution Box wiring & Earthing check.
        - Physical panel cleaning verification.
        - Inverter firmware & Wi-Fi logger check.
    - **Before / After Photo Capture:** Mandatory upload of clean panel photo and inverter running status screen.
    - **Customer Signature Pad:** On-screen digital signature field for customer sign-off.

Module 8: Billing, Invoices & Payment Gateway

8.1 Invoices & Financial Summary Screen

- **Functionality:** Centralized ledger for all financial transactions.
- **Feature Breakdown:**
    - **Invoice Categorization:** Filter by Installation Balance, AMC Subscriptions, Spare Part Replacement, Routine Cleaning Fees.
    - **Invoice Details & Breakdown:** Itemized Cost, HSN/SAC Codes, CGST/SGST/IGST breakdown, Applied Discounts.
    - **PDF Invoice Download:** Generate tax-compliant PDF invoices.

8.2 Digital Payment Integration Screen

- **Functionality:** Multi-channel instant payment collection.
- **Feature Breakdown:**
    - **Payment Methods:** UPI (Google Pay, PhonePe, Paytm), Credit/Debit Cards, Net Banking, EMI Options.
    - **Payment Receipt & Auto-Ledger Adjustment:** Immediate updating of invoice status to Paid with digital receipt generation.
    - **Partial Payment Logic:** Allows milestone payments (e.g., 20% advance, 70% material delivery, 10% post-commissioning).

Module 9: Dynamic Master & Dropdown Configuration Engine

9.1 Unified Dynamic Dropdown System

- **Functionality:** Core architecture requirement where ALL dropdown selections across the application are fed by a single generic API endpoint (fn_get_dropdown_data).
- **Feature Breakdown:**
    - **Single Source Config:** State lists, City lists, DISCOM names, Complaint Categories, Equipment Brands, and Service Types managed via centralized database reference tables (ref_data_categories and ref_data).
    - **Cascading / Dependent Dropdowns:**
        - Selecting a State (e.g., *Maharashtra*) dynamically filters the District/City dropdown without app updates.
        - Selecting a District dynamically populates the local Electricity DISCOM branches (e.g., *MSEDCL*, *Tata Power*).
- **Business Rules:**
    - All dropdown responses must be cached locally in the Flutter app state to allow smooth offline-first dropdown rendering.

Module 10: System Logging, Sessions & Audit Trail

10.1 Centralized Database Logging & Audit Engine

- **Functionality:** Full observability of system interactions and exception tracking.
- **Feature Breakdown:**
    - **API & SQL Execution Logging:** Every API call passes api_endpoint and logged_in_user_id into database logging tables (log_api_execution, log_sql_functions).
    - **User Session Auditing:** Captures user login history, IP address, device model, and active session tokens in user_session_details.
    - **Graceful Exception Mapping:** All database execution errors return standardized JSON payloads, enabling friendly error UI popups (e.g., *"System under scheduled maintenance, please retry in 5 minutes"* instead of raw crash screens).

3. Implementation Roadmap & Module Breakdown

To maintain an agile execution flow, modules can be picked up sequentially for implementation in the following order:

Plaintext

Phase 1: Foundation Setup

├── Module 10: System Logging & Session Tables

├── Module 9: Dynamic Dropdown Engine (ref_data)

└── Module 1: Authentication & User Details

Phase 2: Core Customer & Onboarding Pipeline

├── Module 2: Solar Calculator & Lead Management

├── Module 3: Customer Profile & KYC Document Upload

└── Module 4: Asset & Equipment Registry

Phase 3: Service Engine & Technician Workflow

├── Module 5: Preventive Service & AMC Management

├── Module 6: Complaint Ticketing System

└── Module 7: Field Technician Mobile App

Phase 4: Financials & Admin Controls

└── Module 8: Invoices, Billing & Payment Gateway

4. Acceptance Criteria Checklist for Implementation

When picking each module for development, verify against the following universal standards:

1. **Audit Compliance:** Does every created table include the 5 mandatory audit columns (active_ind, created_by, updated_by, created_on, updated_on)?

2. **Business Logic Location:** Is all calculation and transaction logic residing inside PL/pgSQL functions (fn_create_<>, fn_update_<>)?

3. **Data Protocol:** Does the function accept a JSONB input containing api_endpoint and logged_in_user_id and return a structured success/error JSONB payload?

4. **Error Handling:** Are exceptions caught and logged in log_sql_functions without crashing the application?

5. **UI Rendering:** Is the Flutter screen wrapped in SafeArea, handling loading, error, and empty states gracefully with responsive layout spacing?